Causal Inference
Lecture 8

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Basic notations

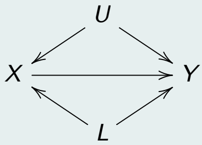
- Exposure: X, numerical (binary or continuous)
- Outcome: Y, numerical, measured at one specific time point.
- Instrumental variable: Z, numerical (scalar or vector).
- Measured covariates: L, scalar or vector (or an empty set). Possible confounders for X and Y
- Unmeasured confounders: U.

The target parameter

- Our goal is to identify and estimate the causal effect of X on Y.

Sufficient for confounding control

Dealing with confounders in observational studies



- In observational studies, we usually don't have exchangeability because we cannot ensure that the measured covariates are sufficient for confounding control.
- In principle, we may achieve conditional exchangeability by controlling for an appropriate set of covariates:

$$(Y_0, Y_1) \perp\!\!\!\perp X | U, L.$$

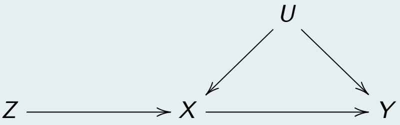
- In practice, the measured covariates L are insufficient for confounding control.

The need for IV

- When exposure is not randomly assigned to participants, the causal effect of the exposure cannot be recovered from simple regression methods.
- A special type of variable called Instrumental can be used to recover the causal effect of the exposure on the outcome even in non-randomised settings.
- IV methods is a collection of methods that can be used to estimate causal effects in the presence of unmeasured confounders U.

Valid instrument

- When U is unmeasured, one can still identify the causal effect of interest by using a special variable Z.
- Such a variable is associated monotonically with the exposure X (relevance), affects the outcome Y only through the exposure (exclusion), and is not confounded with the outcome (exogeneity).
- These three properties define a valid instrumental variable (IV). Such variables allow for the identification of causal effects in the presence of unmeasured confounders U.



DAG of a valid instrumental variable Z

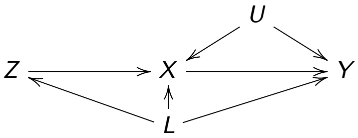
Instrumental variable

- Instrumental variable (IV) regression is a tool that is commonly used in analysis of observational data. Instrumental variables are used to make causal inference about the effect of a certain exposure in a presence of unmeasured confounders.
- A valid instrumental variable is a variable that is associated with the exposure, affects the outcome only through the exposure (exclusion criterion), and is unconfounded with the outcome (exogeneity).
- The IV assumptions are generally untestable and rely on subject-matter knowledge.
- We propose a new method of sensitivity analysis of the G-estimators to invalid instrumental variables. The new method is suitable for linear and non-linear models (specifically, logistic model), and requires only one sensitivity parameter both for the exclusion and the exogeneity assumptions.

Instrumental variable - Introduction

Directed acyclic graph (DAG) of a valid instrumental variable

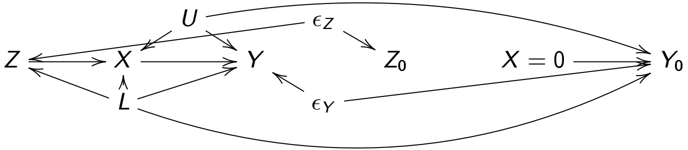
- Let Y be the outcome variable, X the exposure, and Z the instrument. U represents all unmeasured confounders of X and Y, whereas L represents all measured confounders.
- The instrument Z affects Y only through the exposure X, and is unaffected by the unmeasured variables U.



DAG of a causal model with a valid instrumental variable.

Counterfactuals and the twin causal network

- A counterfactual implication of the exogeneity and exclusion assumptions is that a valid IV Z satisfies $Y_0 \perp\!\!\!\perp Z | L$.
- If Z is a valid IV, there is no open path between Y_0 and Z, conditionally on L.



DAG of a twin network causal model with a valid instrumental variable. The left-hand side of the DAG represents the actual world, while the right-hand side represents the hypothetical potential world. $X = 0$ represents the exposure X that is set to 0, and Z_0 represents the potential value of Z in this hypothetical setting.

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A causal model

- Let Y_x be the potential outcome for a given subject with $X = x$.
- Let $m(L)$ be a vector function of the measured variables L.
- Let Z be the instrumental variable.
- Let ξ be the link-function.
- The implicit causal model is given by

$$\xi\left(\mathbb{E}[Y_x|L, X = x, Z, U]\right) = \xi\left(\mathbb{E}[Y|L, X = x, Z, U]\right) = m^T(L)x\psi + \gamma L + f(U)$$

  where the parameter ψ measures the causal effect of the particular exposure level $X = x$ (among those who actually received the exposure level a), conditional on L, Z, and U.
- For identifiability, we assume that the causal effect is constant across instrument levels Z.

Generalized structural mean model - overview

- The generalized structural mean models (GSMM) are semiparametric two-stage models that can handle, among other settings, binary outcomes.
- The first stage model is the generalized structural model indexed with causal parameters that convey the causal effects of interest.
- The second stage model is the association generalized model for the observed data that is required to identify the causal parameters of the former.
- The structural and association models' two link functions can differ. However, possible issues are introduced in such a case when there are no unmeasured confounders.
- Notably, as the association model models the observed data, all its features can be empirically tested. At the same time, the structural model is a latent model defined using counterfactuals. Thus, its features cannot be empirically tested without additional restricting assumptions.

Generalized structural mean model

Formal definition and abuse of notation

- A generalized structural mean model (GSMM) is

$$\xi(\mathbb{E}[Y_1|Z, L, X = x]) - \xi(\mathbb{E}[Y_0|Z, L, X = x]) = m_x^T(L)x\psi_x,$$

  where ξ is a link function of a generalized linear model, ψ_x is the vector of causal parameters, and $\dim(m_x(L)) = \dim(\psi_x)$. The composition of $m_x(L)$ defines the exact form of the causal model.
- We exclude U from the condition, since a valid IV allows for identification of the causal parameters.
- The conditioning on the instrument Z is "abuse of notation" as it not simply conditioning, but denote the utilization of Z in order to identify the causal parameters (e.g., TSLS or G-estimation)
- Assume that the association model of the observed outcome follows a parametric structure indexed by the vector of coefficients β_Y, i.e., $\xi(\mathbb{E}[Y|Z, L, X; \beta_Y])$. Such models are referred to as double ξ models.

Conditioning on the instrument

- A generalized structural mean model is

$$\xi(\mathbb{E}[Y_x|L,Z,X=x]) - \xi(\mathbb{E}[Y_0|L,Z,X=x]) = m_x^T(L)x\psi_x.$$

- Although the unmeasured confounders are implicit, the consequence of their omission is incorporated in the values of β_Y on the observed outcome model $\xi(\mathbb{E}[Y|Z,L,X;\beta_Y])$. Specifically, controlling for the exposure X in the setting with unmeasured confounders U opens a non-causal path between Z and the observed outcome Y and the potential outcome Y_x.
- Formally, X is a collider, thus, conditioning on X opens the non-causal paths

$$Z \to X \leftarrow U \to Y \quad \text{and} \quad Z \to X \leftarrow U \to Y_x,$$

  therefore, Z appears in the the outcome model and in the causal latent above.
- Since $\xi(\mathbb{E}[Y_x|L,Z,X=x]) = \xi(\mathbb{E}[Y|L,Z,X=x])$ this part is identifiable from the observed model.
- $\xi(\mathbb{E}[Y_0|L,Z,X=x])$ is counterfactual, therefore its identification relies on the availability of a valid IV Z.

The main idea - estimation

- A generalized structural mean model is

$$\xi(\mathbb{E}[Y_x|L, Z, X = x]) - \xi(\mathbb{E}[Y_0|L, Z, X = x]) = m_x^T(L)x\psi_x.$$

- The underlying idea of the G-estimation approach is to combine the structural and association models to construct a prediction of the counterfactual outcome for each subject. Then, this prediction and an the auxiliary function of the instrument Z is used to construct estimating equations.
- The values of the causal parameters that solve these estimating equations are called the G-estimators.
- Loosely speaking, the G-estimators are the values of the causal parameters ψ under which a valid IV assumption $Y_x \perp\!\!\!\perp Z|L$ holds in the observed data.
- Note: in linear models there is no need in the observed outcome model. Namely, the latent causal model and a valid IV are suffice for identification and estimation of the causal parameter.

The general form of $h(\psi)$ for non-linear link-functions

- The G-estimator is the value of the causal parameter ψ under which the assumption $Y_0 \perp\!\!\!\perp Z|L$ holds.
- The potential outcome Y_0 is counterfactual for $x \neq 0$, whose mean is estimated by $h(\psi)$. For a link function ξ, the form of $h(\psi)$ is

$$h(\psi) = \xi^{-1}\left(\xi\left(E[Y|X,Z,L]\right) - m^T(L)X\psi\right).$$

- Special cases of $h(\psi)$ for linear, log, and logit link functions, respectively.

$$h(\psi;\alpha) = \begin{cases} Y - m_T(L)x\psi, \\ Y \exp\{-m_T(L)x\psi\}, \\ \text{expit}\left(\text{logit}\left(\mathbb{E}[Y \mid X = x, Z, L] - m_T(L)x\psi\right)\right), \end{cases}$$

G-estimation - formal definition

Estimating equations

- Let $h(\psi)$ be the residual where the causal effect is removed from the factual outcome Y.
- Therefore, $h_i(\hat{\psi})$ as an estimator of the counter-factual value Y_0, i.e., $\hat{Y}_{0i} = h_i(\hat{\psi})$.
- Since $Y_0 \perp\!\!\!\perp Z|L$ implies that IV assumptions are satisfied, we require $d(L,Z) \perp\!\!\!\perp Y_0|L$. Thus the value of ψ that solves the following estimating equation is the G-estimator of ψ

$$H(\psi) = \sum_{i=1}^{n} \hat{d}(L_i, Z_i) h_i(\psi) = 0,$$

  where $d(L,Z)$ is arbitrary function such that $\mathbb{E}[d(L,Z)|L] = 0$ (more details below).
- The main idea: G-estimator is an estimator which makes the potential outcome Y_0 independent of the instrument Z in the sample, conditionally on the measured covariates L.

The $d(L, Z)$ function

The function $d(L, Z)$ is arbitrary function with the same dimension as ψ, such that $\mathbb{E}[d(L, Z)|L] = 0$. The `ivtools` package allows for two choices of $d(L, Z)$

1. $d(L, Z) = m(L)(Z - \mathbb{E}[Z|L])$
2. $d(L, Z) = m(L)(\mathbb{E}[X|L, Z] - \mathbb{E}[X|L])$.

- The first function requires to model only the instrument-nuisance model $\mathbb{E}[Z|L]$, whereas the second function requires both $\mathbb{E}[X|L, Z]$ and exposure-nuisance model $\mathbb{E}[X|L]$, where, generally, the link-function is not preserved under marginalization. That is, for non-linear exposure models, the second function may require two distinct model.
- The first function requires the instrument Z to be a scalar variable, whereas the second function allows for Z to be a vector of arbitrary length.

Assumptions

- Unbiased estimating equations imply, under a valid IV, consistent G-estimators. Hence, we prove the consistency of the G-estimator by proving the unbiasedness of the estimating equations w.r.t 0.
- We prove it for the linear model, i.e., identity link-function. The generalization for non-linear link function is straight-forward.
- Recall the assumption $\mathbb{E}[d(L,Z)|L] = 0$, and the causal model

$$\mathbb{E}[Y|X,Z,L] - \mathbb{E}[Y_0|X,Z,L] = m^T(L)X\psi,$$

and thus we define $h(\psi) = Y - m^T(L)X\psi$.

Proof

$$\begin{aligned}\mathbb{E}[d(L,Z)h(\psi)] &= \mathbb{E}[\mathbb{E}[d(L,Z)(Y - m^T(L)X\psi)|X,L,Z]] \\ &= \mathbb{E}[d(L,Z)(\mathbb{E}[Y|X,L,Z] - m^T(L)X\psi)]\end{aligned}$$

plugging in the aforementioned causal model instead of $m^T(L)X\psi$, we get

Proof - cont.

$$\begin{aligned}
\mathbb{E}[d(L,Z)h(\psi)] &= \mathbb{E}[d(L,Z)(\mathbb{E}[Y|X,L,Z] - (\mathbb{E}[Y|X,L,Z] - \mathbb{E}[Y_0|X,L,Z]))] \\
&= \mathbb{E}[d(L,Z)\mathbb{E}[Y_0|X,L,Z]] \\
&= \mathbb{E}[\mathbb{E}[d(L,Z)Y_0|X,L,Z]] \\
&= \mathbb{E}[d(L,Z)Y_0] \\
&= \mathbb{E}[\mathbb{E}[d(L,Z)Y_0|L]],
\end{aligned}$$

using the fact that Z is a valid instrument, i.e., $Y_0 \perp\!\!\!\perp Z|L$, we have

$$\begin{aligned}
\mathbb{E}[d(L,Z)h(\psi)] &= \mathbb{E}[\mathbb{E}[d(L,Z)|L]\mathbb{E}[Y_0|L]] \\
&= \mathbb{E}[0\mathbb{E}[Y_0|L]] \\
&= 0,
\end{aligned}$$

where the second inequality stems from $\mathbb{E}[d(L,Z)|L] = 0$. $\square$

Consistency, distribution and variance

- The G-estimation details derived from the M-estimation calculus (a.k.a. The generalized estimating equations - GEE).
- The G-estimators are consistent under a valid IV.
- The asymptotic distribution of the G-estimators is approximately normal.
- The asymptotic covariance matrix can be computed using the sandwich formula. Therefore, analytical confidence intervals can be constructed.

Asymptotics of the G-estimators

- The asymptotic variance of θ's estimators $\hat{\theta}$ is obtained by

$$\text{Var}(\hat{\theta}) = n^{-1}A(\theta)^{-1}B(\theta)A(\theta)$$

where the "bread" matrix is $A(\theta) = \mathbb{E}[-\partial Q(\theta)/\partial\theta^T]$, the "meat" matrix is $B(\theta) = \mathbb{E}[Q(\theta)Q(\theta)^T]$, and $Q(\theta)$ is a shorthand for the vector of the estimating function.

- The asymptotic distribution of the estimators $\hat{\theta}$ is multivariate normal

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{D} N_p(0, \text{Var}(\theta)),$$

where the subscript p denotes the dimension of the parametric space $\theta \in \Theta$. For the sample version of "bread" $A(\theta)$ and the "meat" $B(\theta)$ matrices, we replace the expectation operator with the corresponding sample means, and θ with its estimator $\hat{\theta}$.

Linear model

Let X be a binary exposure, and L and Z be scalar variables then assume

1. A causal linear model
$$\mathbb{E}[Y_1|L, Z, X = 1] - \mathbb{E}[Y_0|L, Z, X = 1] = m^T(L)X\psi,$$
such that $m^T(L) = (1, L)$, thus $m^T(L)X\psi = X\psi_0 + XL\psi_1$.
1. An instrument model
$$\mathbb{E}[Z|L] = \theta_3 + \theta_4 L.$$

Now we can define the functional elements of $H(\psi)$

3. $d(L, Z) = (Z - \theta_3 - \theta_4 L, L(Z - \theta_3 - \theta_4 L))^T$
4. $h(\psi) = Y - m^T(L)X\psi = Y - X\psi_0 - XL\psi_1$,

next we construct the estimating equations w.r.t. $\psi = (\psi_0, \psi_1)^T$.

The system of equations

Let $\{(Y_i, X_i, L_i, Z_i)\}_{i=1}^n$ be an iid sample of size n, then system of equations is

$$\sum_{i=1}^{n} \begin{bmatrix} (Z_i - \theta_3 - \theta_4 L_i) \\ L_i(Z_i - \theta_3 - \theta_4 L_i) \\ (Z_i - \theta_3 - \theta_4 L_i)(Y_i - x_i\psi_0 - X_i L_i \psi_1) \\ L_i(Z_i - \theta_3 - \theta_4 L_i)(Y_i - x_i\psi_0 - X_i L_i \psi_1) \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

- If this system has a unique solution $(\hat{\theta}, \hat{\psi}) \in \mathbb{R}^4$, then $\hat{\psi} = (\hat{\psi}_0, \hat{\psi}_1)^T$ is the G-estimator of the mean causal effect in the aforementioned linear model.
- Although these linear estimating equations can be solved analytically, for non-linear link functions ξ, the solution is usually obtained by numerical methods.
- Note that in linear models no explicit specification of the observed outcome model is required.

Causal effect in linear model

Assume that ξ is the identity link function, and that the exposure X is binary, then

$$\mathbb{E}[Y|X=1,L,Z,U]-\mathbb{E}[Y_0|X=1,L,Z,U]=m(L)^T\psi.$$

- If $m(L)=1$ then the exposure effect is just the parameter of interest itself ψ, which represents the effect of increasing the exposure level X from 0 to 1, given L, Z and U, and $A=1$.
- In the general case of $m(L)^T X\psi$, the vector valued function $m(L)$ allows for interaction between the exposure X and the measured variables L.
- To allow for the main causal effect of the exposure X on the outcome Y, the first element of $m(L)$ is typically set to 1.

Linear model

- Assume that following "regular" (i.e., w.r.t the observed variables) model where $m(L) = 1$

$$Y = \gamma + \psi X + f(U) + \epsilon = \gamma + \psi X + \epsilon_u$$

where ψ is the causal effect of X on Y, and $f(U)$ is some function of the unmeasured confounders. Since $f(U)$ is a confounding variable, then $\mathbb{E}[\epsilon_u|X] = \mathbb{E}[f(U)|X] \neq 0$ and since it depends on X, it cannot be absorbed into the intercept term.

- Using the aforementioned notations, one can express this causal model as

$$\mathbb{E}[Y|X = x, L, Z] - \mathbb{E}[Y_0|X = x, L, Z] = \psi x.$$

Linear structural equation model with an IV

Assume we obtain a valid instrument Z that satisfies the three assumptions. Hence, before estimating the OLS of ψ, we first estimate the following linear model (stage one)

$$X = \gamma_0 + \gamma_1 Z + \epsilon_x$$

for the second stage, we regress Y on $\hat{X}$ from stage one instead on X, i.e.,

$$Y = \gamma + \psi \hat{X} + \epsilon,$$

then the OLS estimate of ψ is

$$\psi_{TSLS} = \frac{\widehat{cov}(\hat{X}, Y)}{\widehat{Var}(\hat{X})}.$$

The idea is to replace the observed exposure with the predicated exposure. Since the predicted exposures are obtained from the IV model, they mimic the exposure that would have been observed in a controlled randomised trial.

Consistency of the two-stage estimator

Consistency of the TSLS estimate of ψ

$$\begin{aligned}\psi_{TSLS} = \frac{\widehat{cov}(\hat{X}, Y)}{\widehat{Var}(X)} &\xrightarrow{p} \frac{cov(\hat{X}, Y)}{Var(X)} = \frac{cov(\gamma_0 + \gamma_1 Z, Y)}{Var(\gamma_0 + \gamma_1 Z)} \\ &= \frac{\gamma_1 cov(Z, Y)}{\gamma_1^2 Var(Z)} = \frac{Var(Z)}{cov(X, Z)} \cdot \frac{cov(Z, \gamma + \psi X + \epsilon_u)}{Var(Z)}\end{aligned}$$

using the independence of Z and U, we obtain

$$\psi_{TSLS} \xrightarrow{p} \psi \frac{Var(Z) cov(Z, X)}{cov(Z, X) Var(Z)} = \psi.$$

Equivalence of the G-estimator and the two-stage estimator

Assume a the following causal linear model

$$\mathbb{E}[Y_1|L, Z, X = 1] - \mathbb{E}[Y_0|L, Z, X = 1] = m^T(L)X\psi,$$

such that $m(L) = 1$. Therefore,

$$h_i(\psi) = Y_i - \psi X_i,$$

and $d(L_i, Z_i) = Z_i - \bar{Z}_n$. Hence the estimating equation is

$$\sum_{i=1}^{n}(Z_i - \bar{Z}_n)(Y_i - \hat{\psi}X_i) = 0,$$

after simple algebra, we obtain

$$\psi_G = \frac{\sum_{i=1}^{n} Y_i Z_i - n\bar{Y}\bar{Z}}{\sum_{i=1}^{n} X_i Z_i - n\bar{X}\bar{Z}} = \frac{\widehat{cov}(Y, Z)}{\widehat{cov}(X, Z)},$$

which is the Wald ratio and thus equals the two-stage estimator of ψ.

TSLS estimator example summary

TSLS properties

- The consistency of ψ_{TSLS} is independent of the specified model for the exposure X. Namely, we get an asymptotically unbiased estimator even if the exposure-nuisance model $\mathbb{E}[X|L,Z]$ is misspecified. However, the fitted model $\mathbb{E}[X|L,Z]$ has to be linear (even if it is incorrect w.r.t. true model).
- The method produces a consistent estimator of the conditional causal effect ψ where there are both measured L and unmeasured confounders U.
- The two-stage estimator is a consistent estimator of ψ where the causal model is linear (i.e., for the identity link function ξ).
- For non-linear link-functions ξ this method results in inconsistent estimator of ψ even if the effect measures are collapsible.
- The TSLS is a special case of a G-estimators for linear causal model.

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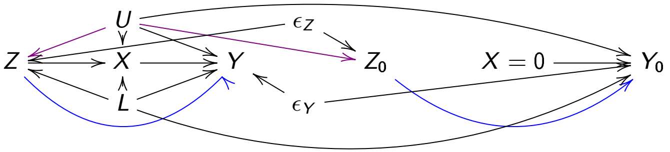
- The blue arrow from Z to Y represents the exclusion criterion violation, i.e., a direct effect of the instrument on the outcome. The violet arrow from U to Z represents represents the exogeneity assumption violation. $X = 0$ represents the exposure X that is set to 0, and Z_0 represents the potential value of Z in this setting.
- Since Z is not influenced by X, thus $Z_0 = Z$, therefore the assumption $Y_0 \perp\!\!\!\perp Z|L$ is violated by $Z_0 \to Y_0$,



DAG of a twin network causal model with invalid instrumental variable. The left-hand side of the DAG represents the actual world, while the right-hand side represents the hypothetical potential world.

Exclusion assumption violation

- Discussion based on notations and definitions from Wang et al. (2018).

Potential outcomes

$Y(x, z)$ is the potential outcome when exposure X is set to x and IV Z to z.

Direct Causal Effect

Direct causal effect of the IV Z on the outcome Y, $Z \to Y$,

$$Y(0, z) - Y(0, 0) = \delta_1 z$$

- Assumes homogeneity of the direct causal effect of IV Z on outcome Y across individuals.

This definition implies also

Formulation

$$\mathbb{E}\left[Y(0, Z) - Y(0, 0) \mid Z\right] = \delta_1 Z.$$

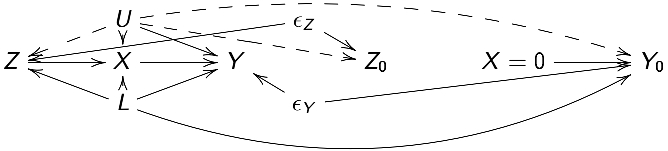
Definition by Wang et al. 2018

Induced Association by Unmeasured Confounders,

$$Z \longleftarrow U \longrightarrow Y$$

$$\mathbb{E}\left[Y(0,0) \mid Z=z\right] - \mathbb{E}\left[Y(0,0) \mid Z=0\right] = \delta_2 z$$

The unmeasured confounder U induces an association between IV Z and outcome Y.



DAG of a twin network causal model with invalid instrumental variable. The left-hand side of the DAG represents the actual world, while the right-hand side represents the hypothetical potential world.

Generalization to Link Functions

- Wang et al. (2018) considered a model with an identity link function ξ.
- We extend their definitions to the logit link function. It allows expressing violations on the logit scale.

Exclusion assumption violation

$$\operatorname{logit}\mathbb{E}\left[Y(0,Z) \mid Z\right] - \operatorname{logit}\mathbb{E}\left[Y(0,0) \mid Z\right] = \delta_1 Z$$

Exogeneity assumption violation

$$\operatorname{logit}\mathbb{E}\left[Y(0,0) \mid Z\right] - \operatorname{logit}\mathbb{E}\left[Y(0,0) \mid Z=0\right] = \delta_2 Z$$

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The G-estimator

- The consistency of the G-estimator relies on the validity of the instrument Z.
- The violation of the conditional independence of Y_0 and Z can arise either from the violation of the exclusion, the exogeneity assumption, or both.

Sensitivity parameter α

- We model compositions of mean independence violations using a parametric function $b(L, Z; \alpha)$, such that

$$\xi(E[Y_0|L, Z]) = a(L) + b(L, Z; \alpha).$$

- In the function $b(L, Z; \alpha)$, the parameter α is a sensitivity parameter that incorporates the conditional association between Y_0 and Z caused by the violation(s).
- Particularly, we require $b(L, Z; 0) = 0$, and $b(L, Z; \alpha) \neq 0$ for $\alpha \neq 0$.
- For any nonzero value of α, the G-estimator is inconsistent.

Consistency of the G-estimators for invalid IVs

- To ensure the consistency of the G-estimators, we reformulate $h(\psi)$ as a function of α

$$h(\psi;\alpha) = \xi^{-1}\left(\xi(\mathbb{E}[Y|X,Z,L]) - m^T(L)X\psi - b(L,Z;\alpha)\right).$$

- For the true α, α^*, in $h(\psi;\alpha^*)$, the G-estimator is consistent.
- The true parameter α^* is non-identifiable and in real-world applications is rarely known.
- A sensitivity analysis is carried out by varying α over a range of plausible values, and mapping each value to a corresponding G-estimator.

$h(\psi)$ as a function of the parameter α

$h(\psi)$ function with sensitivity parameter α

$$h(\psi;\alpha) = \begin{cases} Y - m_T(L)x\psi - b(L,Z;\alpha), \\ Y\exp\{-m_T(L)x\psi - b(L,Z;\alpha)\}, \\ \operatorname{expit}\left(\operatorname{logit}\left(\mathbb{E}[Y \mid X = x, Z, L] - m_T(L)x\psi - b(L,Z;\alpha)\right)\right), \end{cases}$$

$$\mathbb{E}\left[h(\psi;\alpha)\,\middle|\,Z,L\right] = \mathbb{E}\left[h(\psi;\alpha)\,\middle|\,L\right] = a(L), \quad \text{for the true value of } \alpha \text{ denoted by } \alpha = \alpha^*$$

- For α^*, the assumption $Y_0 \perp\!\!\!\perp Z \mid L$ is still violated; however, the estimating functions remain unbiased, and thus the G-estimator is consistent with respect to the true causal parameter ψ.
- The consistency of the G-estimator depends on the correct specification of α in $b(L,Z;\alpha)$. Therefore, we define α as a **sensitivity parameter**.
- The exact value of α^* is typically not known to the analyst.
- A **sensitivity analysis** can be carried out by:
  - Varying α over a range of values considered plausible by subject-matter experts.
  - Estimating the causal exposure effect ψ separately for each value of α.

Proof equivalence between the two definitions

- Wang et al. (2018) defined the exogeneity assumption violation as:

$$\mathbb{E}\left[Y(0,0) \mid Z = z\right] - \mathbb{E}\left[Y(0,0) \mid Z = 0\right] = \delta_2 z$$

and the exclusion assumption violation as:

$$\mathbb{E}\left[Y(0,Z) \mid Z\right] - \mathbb{E}\left[Y(0,0) \mid Z\right] = \delta_1 z\,.$$

- For a generalized linear model with link function ξ, we define the violation as:

$$\xi\left(\mathbb{E}[Y_0 \mid Z]\right) - \xi\left(\mathbb{E}[Y_0 \mid Z = 0]\right) = \alpha Z\,.$$

- Proof of the equivalence of the two definitions

$$\begin{aligned}
&\xi\left(\mathbb{E}[Y_0 \mid Z]\right) - \xi\left(\mathbb{E}[Y_0 \mid Z = 0]\right) \\
&= \xi\left(\mathbb{E}[Y(0,Z) \mid Z]\right) - \xi\left(\mathbb{E}[Y(0,0) \mid Z = 0]\right) \\
&= \xi\left(\mathbb{E}[Y(0,Z) \mid Z]\right) - \xi\left(\mathbb{E}[Y(0,0) \mid Z]\right) + \xi\left(\mathbb{E}[Y(0,0) \mid Z]\right) - \xi\left(\mathbb{E}[Y(0,0) \mid Z = 0]\right) \\
&= \delta_1 Z + \delta_2 Z \\
&= \delta Z
\end{aligned}$$

where we can denote $\delta = \alpha$.

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Mendelian randomization

- The underlying principle of the method is to use genotypes as IVs to estimate the causal effect of phenotype on disease-related outcomes.
- The population distribution of genetic variants is assumed to be independent of behavioural and environmental factors that usually confound the effect of exposure on the outcome.
- The process governing the distribution of genetic variants in the population resembles the randomisation mechanism in RCTs.

Vitamin D data description

- The phenotype is vitamin D status, the outcome is survival at the follow-up endpoint, and the genotype is mutations in the filaggrin gene.
- These mutations are associated with a higher serum vitamin D concentration.
- We used the modified version of data obtained in the Monica10 population-based study. This is a 10-years follow-up study that started in 1982-1984 and initially included an examination of $3,785$ individuals of Danish origin.

Data description

- We use a data set `VitD` borrowed from Martinussen et al. (2018) on a cohort study of vitamin D and mortality.
- We will estimate the causal effect of vitamin D on mortality by IV analysis, using mutations in the filaggrin gene as an instrument.
- These mutations have been shown to be associated with higher vitamin D status (Skaaby et al. 2013).
- The `VitD` data frame contains 2571 subjects and the following 5 variables:
  1. age (at baseline) - A measured covariate (confounder) L.
  2. filaggrin (binary indicator of mutations presence) - An instrument Z.
  3. vitd (vitamin D level) - The exposure X.
  4. time (follow-up time) - The time-to-event outcome T.
  5. death (a binary indicator of a subject's death) - The binary point outcome Y.

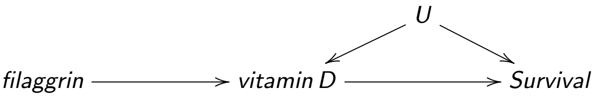
Background on vitamin D effects

- Vitamin D deficiency has been linked with several lethal conditions such as cancer and cardiovascular diseases.
- Vitamin D status is also associated with unmeasured behavioral and environmental factors that may result in biased estimators when using standard statistical analyses to estimate causal effects.
- Mutations in the filaggrin gene have been shown to be associated with a higher vitamin D status and are assumed to satisfy the IV assumptions.
- We use a publicly available mutilated version of the data on a cohort study on vitamin D status causal effect on mortality rates.
- The data frame contains 2571 subjects and 5 variables: age (at baseline), filaggrin (a binary indicator of whether filaggrin mutations are present), vitd (vitamin D level at baseline, measured as serum 25-OH-D(nmol/L)), time (follow-up time), and death (an indicator of whether the subject died during follow-up).

The vitamin D example

Data preparation

- We use the binary indicator of survival at the follow-up endpoint as the outcome Y.
- We used the binary indicator of the presence of the filaggrin gene mutation as the IV Z.
- We use the continuous vitamin D level as the exposure X.
- We use the continuous age variable as a measured confounder L.
- The estimated model was the double logit model.
- We discard the covariate age to simplify the calculations.



The vitamin D data: G-estimation for binary outcome

Instrument-nuisance model; $\text{logit}(\mathbb{E}[Z]) = \theta_0$

```
glm(formula = filaggrin ~ 1, family = binomial(link = "logit"),
data = VitD)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.50574    0.07467  -33.56   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1375.6  on 2570  degrees of freedom
Residual deviance: 1375.6  on 2570  degrees of freedom
AIC: 1377.6

Number of Fisher Scoring iterations: 5
```

Observed Outcome model: $\text{logit}(\mathbb{E}[Y|Z,X]) = \beta_0 + \beta_1 Z + \beta_2 X + \beta_3 ZX$

```
glm(formula = death ~ filaggrin * vitd2, family = binomial(link = "logit"),
data = VitD)

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)     -0.79914    0.09230  -8.658  < 2e-16 ***
filaggrin       -0.31022    0.36799  -0.843    0.399
vitd2           -0.16943    0.03825  -4.430 9.42e-06 ***
filaggrin:vitd2  0.02899    0.13819   0.210    0.834
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2803.2  on 2570  degrees of freedom
Residual deviance: 2779.1  on 2567  degrees of freedom
AIC: 2787.1

Number of Fisher Scoring iterations: 4
```

G-estimator of the causal effect ψ

```
ivglm(estmethod = "g", X = "vitd2", fitZ.L = fitZ.L2, fitY.LZX = fitY.LZX2,
data = VitD, link = "logit")

Coefficients:
         Estimate Std. Error z value Pr(>|z|)
vitd2   -1.558      1.546  -1.008    0.314
```

- The G-estimator of the mean causal effect ψ is statistically insignificant.
- Assuming that the instrument Z (filaggrin) is valid, the G point estimator of the causal effect is -1.558, i.e., $\hat{\psi}_G(0) = -1.558$.

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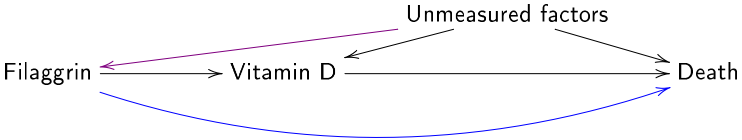


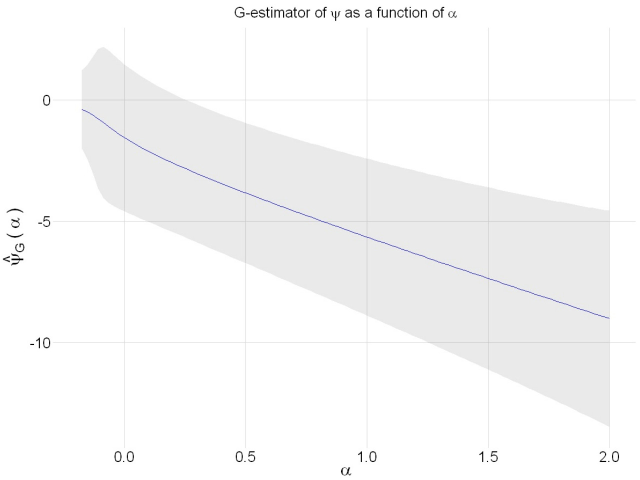
DAG of the vitamin D status model. The violet arrow represents a possible exogeneity assumption violation, and the blue line represents a possible exclusion assumption violation.

Vitamin D data - violation structure

- The assumed violation structure on the logit scale is given by

$$\text{logit}\mathbb{P}(\text{Death}_0 = 1 \mid \text{Filaggrin} = 1) - \text{logit}\mathbb{P}(\text{Death}_0 = 1 \mid \text{Filaggrin} = 0) = \alpha^* Z,$$

where $Z = 1$.



G-estimator $\hat{\psi}_G(\alpha)$ of the vitamin D status causal effect on death rate during follow-up as a function of the sensitivity parameter α.

For $\alpha = 0$ the $\hat{\psi}_G(0) = -1.558$.

For $\alpha < -0.17$ there is no solution to the estimating equations.

Vitamin D data

- If the structural mean model and the association model are correctly specified, there is no evidence of a positive causal effect of vitamin D deficiency on the mortality rate. That is, we can conclude that vitamin D supplements are safe and may either reduce the mortality rate or, in worst case, have no or neglectable positive effects.
- If the IV is invalid, then the true causal effect is likely to be of larger magnitude than the estimated value for $\alpha = 0$.
- The R-code of the analysis is available here:
  `https://github.com/vancak/G_sensitivity/blob/main/Vitamin_D/VitD_logreg`

One parameter sensitivity analysis - advantages

- The one parameter sensitivity analysis approach theoretical framework and practical guidelines on how to conduct sensitivity analysis of G-estimators using single sensitivity parameter that captures violations of both the exogeneity and the exclusion assumptions.
- Using a single parameter is a valuable advantage as fewer parameters are needed to be specified. This feature may increase model stability and decrease the risk of misspecification.
- The proposed method is applicable both to linear and non-linear causal models.
- The one-sensitivity paramter aligns with and generalize the previous work by Wang et al. (2018).

One parameter sensitivity analysis - disadvantages

- The usage of one parameter for two distinct violations complicates its interpretation.
- Specifying a plausible interval for the sensitivity parameter may be challenging since it relies heavily on subject-matter knowledge and a solid understanding of causal inference.

References

- Skaaby, T., Husemoen, L. L. N., Martinussen, T., Thyssen, J. P., Melgaard, M., Thuesen, B. H., ... & Linneberg, A. (2013). Vitamin D status, filaggrin genotype, and cardiovascular risk factors: a Mendelian randomization approach. PloS one, 8(2), e57647.
- Wang X, Jiang Y, Zhang NR, Small DS. Sensitivity analysis and power for instrumental variable studies. Biometrics. 2018;74(4):1150-1160.
- Vancak, V., & Sjölander, A. (2023). Sensitivity analysis of G-estimators to invalid instrumental variables. Statistics in Medicine, 42(23), 4257-4281.